Structured Process for Selecting Components

Table 1

Category	Remarks
TECHNICAL	
Accuracy/Performance of the part	Tolerance of the electronic parts is very critical. Parts like ceramic capacitors have large variation in their tolerances. The same is true for parts which need narrow tolerances. Most of the time engineers make mistakes in the selection of the parts. Using wrong tolerance part in their design.
Alternatives	Next challenge is the alternative part to a part that is used in the design. While equivalents are easy to get for passive parts and some of the active parts like transistors, MOSFETs, SCRs, etc, even if the alternates are available parameter by parameter comparison is a must before the alternate part is cleared for production. But it is nearly impossible in case of microcontrollers/processors. Designers need to spend time to see if they can have alternatives in the same microcontroller family in case the one that is used is not available. This needs planning.
OEM components	Parts like power supplies and display systems are actually finished products by themselves but are components in another product. Choice of the OEM as well as qualification of the product properly is the key element in this case. While choosing a power supply unit (PSU) many designers never bother about their product's need. They just select a PSU and later suffer in the field.
Package type	Most ICs are available in multiple packages. Choice of the package has to be in line with product's design. For example, LED lighting designers use low-cost paper phenolic board, but if you select a QFN part as that may be cheaper or easily available you will run into trouble after soldering. Many PCBs will fail because paper phenolic PCBs warp due to heat and the QFN part's pads will be damaged and the PCBA will fail at testing stage. Packages also impact product manufacturing technology. Many designers tend to pick the smallest footprint for a part without realising that smaller parts are more difficult to assemble even when automated. EMS vendors charge more as the parts become small!
Environment impact	Designers don't choose the right part for the right product working environment. For example, when you select parts for industrial or military applications, components which are qualified for these applications only have to be used. Using normal commercial parts can lead to product failure during environmental testing.
COMMERCIAL	
Lead time	Always choose the part which is either freely available or has a short lead time for delivery. Typically, this is an intensive affair and a proper monitoring process only helps in getting this problem out of the way. This is one critical item which will be used as benchmark in selecting EMS vendors. Good EMS vendors have good mechanism for tracking this.
Fast moving part	When choosing parts, especially controllers and ICs, check with the vendor who part sells in volume. This approach ensures component shortage does not impact as manufacturers always make the fast-moving parts in high volume.
Obsolescence	Another important aspect is the life of the components. Typically, obsolescence hits fairly matured products. Component vendors do give sufficient notice before the part is stopped from production line. However, some of the brand-new ICs are also obsoleted within a year of production due to reasons such as manufacturing yield and lack of sales volume. When choosing a new part, designers need to ensure the part has been in production for at least a year before committing its use to design.
Banned/restricted items	Normally, when new components are introduced, some of them have dual use (for Instance, in normal applications and military use). This is common for the high-end microprocessors. The government of the country where the component originated can restrict the sale only under authorisation, which can be a big issue. While the component vendors normally sound warning about this, things can go in an unexpected way if the government bans export of some dual-use components and to import special permits have to be obtained. This complicates the entire design and other plans. Designers should ensure they choose the right parts which don't fall in this category.
STANDARDS COMPLIANCE	
Environmental compliance RoHS/ REACH	Third category is the need for products to meet environmental compliances. This essentially means the processes used for product manufacturing and assembly have to use environmentally friendly materials. This led to standards like RoHS, REACH, and WEEE. These standards specify the display of material content, use of lead-free (RoHS) soldering and products to be designed such that 98% of It can be recycled and remaining can be safely disposed of (WEEE). This means designers need to use parts which meet these requirements.
Domain qualifications (Auto/Military)	As the electronics content is increasing in automotive and military applications, these components have specific standards (AEC -100 etc, or MIL standard) to comply. These parts are specially processed and qualified in the manufacturing line to meet the standards specified. When products for these applications are used, designers have to select the parts that meet the specifications.
Manufacturing data capture and preservation	This is a key aspect of a component which is needed for some of the applications like medical and avionics devices where adverse incident regulating agencies like FAA & FDA can call for all the data including the components used to do the root cause analysis. If the product is a medical device, typically the manufacturer of the device has to maintain a document called DMR (device master record) which captures component batch number details and all the manufacturing records for future reference.

in a dry environment (in specially designed racks)

7. Similarly, most integrated circuits are sensitive to static electricity. These components have to be stored in static-free area to protect them from getting damaged

8. Equivalent components for replacement (typically applicable for passive resistors and capacitors, and active components like transistors and some logic ICs)

9. Fake parts that are sold as original part but will not work at all

10. Pulled out parts that are pulled out of waste PCBAs and cleaned up and sold as new parts

Why components are critical

Right component selection contributes to more than 60% to a product's performance and reliability. Most electronics parts fail in the early stage of the product's use. Fig. 1 which shows